Shiv Raj Bangad

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Education

Purdue University, Daniels School of Business, Master of Science in Business Analytics and Information Management – West Lafayette, IN — Dec 2026

- **Relevant Courses:** Business Analytics, Data Mining, Machine Learning, Reinforcement Learning, Cloud Computing, Advanced Database Management, MLOps, Spreadsheet Modelling, Computational Finance
- **Certifications:** AWS Certified Cloud Practitioner, AI-900 Microsoft Azure AI Fundamentals

University of Cincinnati, Carl H. Lindner School of Business, BS in Business Analytics and BBA in Finance – Cincinnati, OH — May 2025

- **GPA:** 3.9/4.0 **Honors:** Lindner Business Honors, Kolodzik Business Scholar (top 5% of business students)

Experience

Business Intelligence Engineer Intern, **Amazon (AWS Analytics)** – Seattle, WA — May 2026 - Aug 2026

- Built an AI agent system on AWS Lambda and Bedrock AgentCore that automates end-to-end AWS feature adoption tracking, cutting delivery time from 2 days of manual engineering work to under 30 minutes.
- Engineered scalable PySpark and AWS Glue pipelines on S3 to process 10B+ row, multi-terabyte datasets across 100+ service features, serving as the data layer for agent execution and live dashboards.
- Designed a deterministic AST-based SQL parser & guardrail to validate 100% of LLM-generated SQL against production data contracts, preventing invalid joins, double-counting, and broken schema builds.
- Deployed a containerized MCP (Model Context Protocol) tool server with SigV4 cross-account authentication and a CI/CD pipeline to manage seamless, automated production releases.
- Implemented an NL-to-SQL Q&A engine integrated with z-score baseline statistical anomaly detection to auto-generate weekly executive deep-dive narratives on feature adoption trends.

Financial Analyst Intern, **BDT & MSD Partners (Private Equity)** – Chicago, IL — Jun 2025 - Aug 2025

- Engineered automated cash-flow ETL pipelines with advanced SQL queries and script-driven Excel models, cutting capital workflow processing time by 30% and reducing manual reconciliation effort.
- Automated data validation and schema checks on $100M in capital calls and distributions, ensuring 100% referential integrity across complex institutional investor databases.
- Architected normalized relational data models to consolidate fragmented portfolio metrics across $2B+ AUM, resolving structural data gaps and standardizing historical formats.
- Built a centralized SQL repository that accelerated query throughput and eliminated manual data entry for executive reporting dashboards, streamlining downstream data integration.

Data Scientist Co-op (Cintas), **UC Center for Business Analytics** – Cincinnati, OH — Jan 2025 - May 2025

- Architected an automated Pandas/Dash SKU anomaly detection dashboard with built-in data validation workflows, replacing manual audits and cutting operational review time by 360+ hours annually.
- Built scalable Python data processing scripts using statistical normalization and feature engineering to correct skewed SKU distributions, boosting model predictive performance by 30%.

Data Scientist Co-op (HBH Holdings), **UC Center for Business Analytics** – Cincinnati, OH — Sep 2024 - Dec 2024

- Engineered an automated Python document pipeline using Poppler and Unicode NFKC normalization to extract and structure raw text from complex legal PDF and Word contracts.
- Architected an AI-powered contract Q&A assistant on Haystack and the Mistral 7B LLM, letting non-technical stakeholders query contract terms in plain English with sub-5 second responses.
- Implemented NLP preprocessing and semantic vector indexing, increasing domain-specific search accuracy by 75% and enabling reliable information retrieval across complex legal documents.

Data Analyst Intern, **Great American Insurance Group** – Cincinnati, OH — Apr 2022 - Apr 2024

- Executed and managed monthly data pipeline processing for 70,000+ policies representing approximately $1 billion in total premium value using SQL Server and Access.
- Designed relational Access/SQL Server databases and optimized indexing strategies, reducing analytical data extraction query times by 40% for recurring policy and premium analyses.
- Engineered automated Tableau reporting workflows that replaced legacy manual data preparation, cutting the monthly analytical prep cycle time by 50% for recurring reporting.

Projects

Agentic AI Observability & Platform Advisor, Cummins Aug 2026 - Present

- Architecting an Azure agentic observability cockpit to govern and monitor autonomous AI agents running on Databricks and Snowflake, centralizing visibility into agent behavior and platform usage.
- Developing a RAG-based LLM Platform Advisor that analyzes PySpark jobs and auto-generates CI/CD tests, driving a 50% cost reduction across Databricks and Snowflake workloads.
- Integrating the advisor into CI/CD workflows to automatically validate PySpark code changes before deployment, improving release reliability and governance of agent-generated data pipelines.

Google (YouTube) Capstone Course, Purdue University Jan 2026 - May 2026

- Engineered a distributed PySpark ETL pipeline to ingest, clean, and vectorize unstructured metadata across 1,000+ creator channels, reducing data processing latency by 45%.
- Delivered an end-to-end LLM recommendation platform using RAG and vector embeddings so creators can query cross-channel trends in real time, driving a 15% lift in predicted engagement.
- Built an interactive Dash/Streamlit analytics dashboard to visualize content whitespace and optimal posting cadences across 20+ video categories for data-driven content planning.
- Developed a benchmarking framework using statistical A/B testing simulations to evaluate LLM strategy recommendations against historical performance baselines.

Data4Good National AI Competition Oct 2025 - Jan 2026

- Secured the Regional Championship title by developing a custom DeBERTa-v3 model that achieved 97.5% classification accuracy in the Data4Good National AI Competition.
- Architected an end-to-end NLP framework with automated tokenization, TF-IDF vectorization, and DeBERTa-v3 fine-tuning to classify semantic relationships as factual, contradictory, or irrelevant.
- Delivered a production-ready inference pipeline using a stratified weighted ensemble to aggregate prediction logits, reducing classification error by 15% on unseen evaluation data.

Predictive Modelling for Firm Bankruptcy | Kaggle Competition Nov 2025 - Dec 2025

- Achieved 1st place out of 37 teams with an automated ML pipeline delivering a competition-leading 0.9493 ROC-AUC in corporate bankruptcy prediction on severely imbalanced data.
- Engineered an "Evolutionary XGBoost" feature generation pipeline that synthesized, mutated, and selected non-linear feature interactions across 100 generations to address class imbalance.
- Constructed a 3-stage Stacking Super-Learner integrating feature transformation layers, parallel XGBoost, LightGBM, and CatBoost inference, and a meta-learner execution layer.

Stock Market Return Prediction (Research Assistant) May 2024 - Aug 2024

- Built an automated ingestion pipeline to clean and align macroeconomic indicators and corporate fundamental metrics for quantitative return-prediction analysis.
- Developed feature engineering routines using Principal Component Analysis (PCA) and rolling-window normalization to eliminate multi-collinearity and prevent feature leakage.
- Delivered a multi-model evaluation framework across BART, SVM, and Random Forest, achieving a 10% boost in out-of-sample R^2 prediction accuracy.

Skills

Data Engineering & Analytics: ETL/ELT Pipeline Development, AWS (Glue, EMR, Redshift, S3, Lambda, Bedrock AgentCore, QuickSight), PySpark, Apache Spark, Tableau, Power BI, MySQL, PostgreSQL, SQL Server, Azure, GCP, Hadoop

Languages & Querying: Python (Pandas, NumPy, Scikit-Learn, PyTorch, TensorFlow), SQL (PostgreSQL, MySQL, SQL Server), R, Java, Bash/Zsh

Data Platforms & Storage: Data Warehousing, Relational Databases, NoSQL, Vector Databases (ChromaDB, PGVector), S3 Parquet/Delta Storage, Distributed Storage

Machine Learning & MLOps: DeBERTa, Transformers, LLM Agent Architectures, RAG Systems, XGBoost, LightGBM, Stacking Ensembles, A/B Testing, Feature Engineering, CI/CD, Docker, Git/GitHub